# Validating and Extending a Chapter-Aligned Moodle Engagement Metric Across STAT0002 and STAT0004

## Evidence from online, hybrid and in-person Moodle delivery cohorts

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**Abstract**

Virtual learning environments such as Moodle record when students access course resources, how often they return, and which types of activities they use. These logs offer a scalable route to early student support, but simple click counts are hard to interpret and supervised predictive models typically require historical grade data. This report evaluates a theoretically grounded, chapter-aligned engagement metric that summarises three behaviours for each student: **Frequency** (the number of chapter-specific study sessions completed), **Immediacy** (how promptly a student first accesses newly released chapter material), and **Diversity** (the breadth of distinct resource types used within a chapter). Each component is min-max scaled relative to peers within the same chapter and teaching week, producing scores between 0 and 1, and the three are added to form a chapter-level score, IDF $= F + I + D \in [0, 3]$. Summing chapter scores across all material released by week $t$ gives a cumulative engagement measure that can be updated in real time without outcome data. Building on the metric proposed by Johnston et al. (2025), this report applies the fixed measure to five cohorts from two UCL undergraduate statistics modules: STAT0002 (a first-year probability and statistics course) and STAT0004 (a second-year programming and practical statistics course), together comprising $N = 1{,}291$ students and spanning online, hybrid and in-person delivery periods not used in the original validation. The report asks whether the metric travels to new cohorts, which component carries the most distinct information about final grades, and whether the engagement–grade association is uniform across the grade distribution or concentrated among lower-performing students. Across the pooled cohorts, cumulative IDF shows a moderate positive association with final grade ($r = 0.32$). Once the shared variance among the three components is removed using a relative-importance decomposition, Diversity and Immediacy account for more unique explained variance than Frequency at week 11 (mean shares of 37%, 36% and 27%), even though all three correlate similarly with grade in bivariate analysis. The engagement–grade association is also stronger among lower-performing students: in every cohort, the slope at the 10th percentile of the grade distribution exceeds the median slope by an average of 2.3 grade points per standard deviation of engagement, and indicator-specific quantile models suggest that Immediacy is particularly important in the lower tail. Together, the findings support using the metric for low-cost, supportive monitoring rather than high-stakes prediction. Comparisons across delivery periods remain descriptive because delivery mode is confounded with cohort and academic year.

# 1 Introduction

Student engagement, the time and effort students devote to learning activities, is widely linked to academic performance, satisfaction and retention. In online and blended higher education, part of the *behavioural* dimension of engagement is recorded automatically by virtual learning environments (VLEs): which resources students access, how often they return, and how promptly they engage with newly released material. Moodle logs therefore offer a scalable record of observed study behaviour, but they remain only a proxy. They cannot observe offline study, motivation, attendance or the quality of students' understanding (Henrie et al., 2015).

This creates a practical measurement problem. Simple indicators such as total clicks are easy to compute but difficult to interpret pedagogically; a high count need not represent broad or sustained engagement (Fincham et al., 2019). Threshold systems can flag low activity, but their cut-offs are often arbitrary and may not generalise. Supervised predictive models can be more flexible, yet they require historical outcome data and can fail when moved to new modules, cohorts or delivery contexts (Gašević et al., 2016). For early student support, an attractive alternative is an interpretable metric that is grounded in engagement theory, can be updated during the term, and can be validated without assuming that a low log count is equivalent to low effort.

Johnston et al. (2025) propose such a metric by adapting the earlier course-wide engagement score of Chong and Wong (2019). The original course-wide score combines five indicators after the course has finished. Johnston et al. instead align the measurement with how teaching material is released: for each student, chapter and week, they compute Frequency, Immediacy and Diversity, scale each component relative to peers, and add them into a chapter-level IDF score. Frequency captures the volume of chapter-specific study sessions, Immediacy captures how soon a student first engages with a chapter after it becomes available, and Diversity captures the breadth of resource types accessed. Summing chapter scores over the term gives a cumulative engagement trajectory that can be monitored in real time.

The present report is a secondary validation and extension of that fixed metric, not a new metric-development exercise. It applies the chapter-aligned IDF measure to five cohorts from two UCL undergraduate statistics modules. STAT0002 is a first-year course introducing probability and statistical theory, represented here by an online delivery (2020–21) and an in-person delivery (2022–23). STAT0004 is a second-year programming and practical statistics course covering statistical computing and data analysis, and is the only module spanning all three delivery conditions: online (2020–21), hybrid (2021–22) and in-person (2022–23). These five cohorts were not used in the original validation, providing a genuinely independent test of the metric.

The analysis pursues two substantive questions. The first concerns component importance: once the strong correlations among Frequency, Immediacy and Diversity are accounted for, which component carries the most unique information about final grades? A relative-importance decomposition due to Lindeman et al. (1980), known as the LMG method, addresses this by averaging each predictor's contribution to explained variance over all possible orderings in which the three components could enter a regression model. The second question asks whether the engagement–grade association is uniform across the grade distribution or stronger among students already at academic risk. Quantile regression estimates the slope between engagement and grade separately at different percentiles of the grade distribution, making it possible to ask whether low engagement is more consequential for lower-performing students than for the average student. Supporting analyses assess the overall pooled association, signal timing, engagement trajectories, early-warning precision, assessment-component alignment and robustness to analytic choices. Delivery-period comparisons are treated as descriptive boundary checks rather than causal tests because delivery mode is perfectly confounded with cohort and academic year.

## 2 Background and metric context

Online behavioural engagement refers to observable, sustained interaction with learning activities in a digital environment. Engagement theory distinguishes behavioural, emotional and cognitive forms of involvement; Moodle logs capture primarily the behavioural dimension, participation, effort and attention to academic tasks, rather than motivation or understanding (Fredricks et al., 2004; Henrie et al., 2015; Martin and Borup, 2022). It is only a *proxy* for learning: logs record what happens inside the VLE, not offline study, in-person attendance or the quality of students' reasoning. Even so, when behavioural engagement is measured coherently and validated against academic outcomes, it can support timely monitoring and targeted support.

The course-wide metric of Chong and Wong (2019) combines five indicators (Immediacy, Frequency, Diversity, Recency and Interval) into a single retrospective score aggregated across an entire course. That design is useful for end-of-term analysis but poorly suited to weekly monitoring, and it treats the course as homogeneous rather than aligned with its chapter structure. For real-time monitoring this is problematic: a student may be on schedule with newly released material while having little activity on later chapters that are not yet relevant, or may return frequently to old material without keeping pace with current content.

Johnston et al. (2025) address this by moving from a course-wide score to a chapter-aligned score. At each week $t$, and for each released chapter $k$, three raw indicators are calculated from chapter-specific study sessions: **Frequency**, the number of study sessions associated with the chapter; **Immediacy**, the delay between the chapter's first observed availability and the student's first chapter-specific session, transformed so earlier access receives a higher score; and **Diversity**, the number of distinct Moodle activity or resource types accessed within that chapter. Each raw indicator is min-max scaled across all students for the same chapter $k$ and week $t$, placing it on a common $[0, 1]$ scale relative to peers. For example, the scaled Immediacy indicator is

$$I_{k,t}^{(i)} = \frac{\tilde{I}_{k,t}^{(i)} - \min_i \tilde{I}_{k,t}^{(i)}}{\max_i \tilde{I}_{k,t}^{(i)} - \min_i \tilde{I}_{k,t}^{(i)}},$$

and the same transformation is applied to $\tilde{F}$ and $\tilde{D}$. A student who scores 1 on all three components engaged with the chapter earlier, more frequently, and across a broader range of resource types than any of their peers. The chapter-level IDF score is then the simple sum

$$IDF_{k,t}^{(i)} = F_{k,t}^{(i)} + I_{k,t}^{(i)} + D_{k,t}^{(i)} \in [0, 3],$$

so a high score means that a student engages often, early and broadly with that chapter. Chapter scores are then summed over released chapters to obtain an overall cumulative engagement measure for student $i$ at week $t$. Recency and Interval are excluded from the real-time version because they become unstable when measured before the course has finished.

Chapter alignment matters because engagement is most interpretable when measured in step with how material is released. For early-warning use, practitioners also need to understand the recall–precision trade-off: flagging the lowest-engagement students may capture many who ultimately struggle, but also many who would have succeeded without intervention (Conijn et al., 2017). The present report takes the metric definition as given and focuses on how it behaves on new UCL cohorts and what additional diagnostic insight can be extracted from its components and related temporal patterns.

# 3 Data and engagement measures

The analysis uses five module-year deliveries from two undergraduate statistics modules at University College London. STAT0002 is a first-year foundations module represented here by an online cohort (2020–21) and an in-person cohort (2022–23). STAT0004 is a second-year module and the only module in this dataset spanning all three delivery conditions: online (2020–21), hybrid (2021–22) and in-person (2022–23). Each delivery covers eleven teaching weeks with up to ten labelled chapters. Final grades are recorded on a 0–100 scale; grades below 50 were treated as low performance, and grades below 40 as failure. Students with a recorded grade of zero, typically reflecting exam absence, were excluded.

Table 1 summarises the cohorts. The main engagement–grade sample comprises $N = 1{,}291$ students with both engagement records and valid final grades after exclusions. The latent trajectory analysis below uses $N = 1{,}290$ students with complete weekly engagement profiles suitable for class assignment; the difference of one student reflects missing weekly coverage in the trajectory model, not an error in the cohort inventory.

**Table 1:** Cohort inventory for the main engagement–grade sample ($N = 1{,}291$).

| Module | Year | Condition | $N$ | Mean (SD) | $\% < 50$ | $\% < 40$ |
|---|---|---|---|---|---|---|
| STAT0002 | 2020–21 | Online | 342 | 66.8 (13.8) | 9.9 | 1.8 |
| STAT0002 | 2022–23 | In-person | 268 | 67.4 (13.0) | 10.4 | 2.6 |
| STAT0004 | 2020–21 | Online | 312 | 62.7 (12.2) | 12.5 | 2.9 |
| STAT0004 | 2021–22 | Hybrid | 172 | 71.9 (8.8) | 1.7 | 0.0 |
| STAT0004 | 2022–23 | In-person | 197 | 70.6 (11.3) | 3.6 | 1.0 |
| **Total** | | | **1,291** | | | |

*Source: author's analysis of UCL Moodle log data.*

Two design features constrain interpretation. First, the share of low-performing students varies widely across cohorts (from 1.7% to 12.5%), which limits how sharply any early-warning flag can perform in the highest-achieving deliveries. Second, delivery condition is **perfectly confounded** with academic year and cohort: online corresponds to 2020–21, hybrid to 2021–22, and in-person to 2022–23. Comparisons across conditions therefore describe different module-year contexts rather than randomly assigned delivery modes, and no causal claim about online, hybrid or in-person delivery can be supported.

The modules also differed in assessment structure. STAT0002 in-person records separate exam, participation and peer-marking components; STAT0004 deliveries include coursework and quiz marks, with a group component in the in-person year. This matters because Moodle engagement may align more closely with individual or exam-style assessment than with group, peer-marked or participation components. Where component marks were available, this report therefore analyses whether the engagement metric correlates more strongly with individually assessed work than with collaborative or peer-assessed components.

The two modules also differ in the strength of the engagement–grade link across deliveries. STAT0002 cohorts show end-of-term correlations of $r = 0.28$–$0.41$, whereas STAT0004 ranges from $r = 0.19$ to $r = 0.30$. As interpretive context, the two modules may differ in ways that affect how well Moodle activity proxies study behaviour; several factors may contribute, though they cannot be disentangled in this observational design. STAT0002 is a first-year foundations module with substantial individually assessed exam weight, where sustained access to lecture materials and problem sheets may track preparation more directly. STAT0004 is a second-year module with greater student autonomy, more coursework and group-assessment weighting, and possibly more study away from Moodle. Each of these factors would weaken the observable link between VLE logs and final outcomes. How tightly Moodle resources are organised by chapter may also matter: where chapter labelling aligns less clearly with assessed work, a chapter-aligned metric would legitimately carry less outcome signal, as Johnston et al. (2025) report for less structured courses.

### 3.1 Engagement feature construction

The supplied engagement table contains weekly chapter-level contribution columns for Frequency, Immediacy and Diversity. These are the operational form of the Johnston et al. IDF components used in this report: each column corresponds to a student–week–chapter contribution for one component. Before modelling, the data were checked to clarify their time semantics. A reconstruction from the raw Moodle event log showed that the supplied contribution values behave as *per-week increments*, not as already cumulative totals. This matters because the paper's real-time metric is cumulative up to week $t$; therefore the analysis must first build weekly totals and then accumulate them explicitly.

For each student and week, contribution values were summed across released chapters to obtain weekly Frequency, Immediacy and Diversity totals. The weekly composite was then defined additively as

$$\mathrm{IDF}_{\mathrm{week}} = \mathrm{freq}_{\mathrm{week}} + \mathrm{imm}_{\mathrm{week}} + \mathrm{div}_{\mathrm{week}},$$

and cumulative measures were obtained by summing across weeks: separately for Frequency, Immediacy, Diversity and the composite IDF:

$$\mathrm{cum\_IDF}_{i,t} = \sum_{s=1}^{t} \mathrm{IDF}_{\mathrm{week},i,s}.$$

This additive definition follows the chapter-level formulation of Johnston et al. (2025); it is not a multiplicative composite and no outcome data are used to create the engagement measures. The reconstruction check confirmed that the supplied contributions are most consistent with the intended per-week Frequency and Diversity definitions, while Immediacy is retained as supplied because it depends on first-access timing and release-date assumptions.

For extension analyses, additional timing and regularity features were derived from event logs (for example, variation in weekly activity and gaps between sessions). Latent-class mixed models were also fitted to normalised weekly cumulative IDF curves to identify distinct engagement trajectories over the term.

# 4 Methods

## 4.1 Overall association with final grade

Within each cohort, Spearman rank correlations were computed between cumulative engagement and final grade at each week and at the end of term. To synthesise evidence across cohorts, cohort-specific correlations were pooled using a random-effects meta-analysis implemented with `metafor` (Viechtbauer, 2010), reporting the pooled correlation, confidence interval and $I^2$ heterogeneity statistic. Complementing rank-based evidence, a mixed-effects model was fitted with final grade as the outcome and standardised cumulative IDF as the predictor, allowing the slope to vary by cohort. This expresses the association on the grade scale in points per one standard deviation of engagement.

## 4.2 Component importance: relative-importance decomposition

Because Frequency, Immediacy and Diversity are strongly correlated (mean pairwise $|r| \approx 0.75$–$0.83$; variance inflation factors VIF $\approx 2.6$–$4.4$ at weeks 6 and 11), ordinary partial regression coefficients are unstable under collinearity and can even reverse sign across cohorts. To obtain stable estimates of each component's unique contribution, the Lindeman–Merenda–Gold (LMG) method (Lindeman et al., 1980) is used. LMG partitions the model $R^2$ from the regression of final grade on Frequency, Immediacy and Diversity into additive shares, one per predictor, by averaging each predictor's marginal contribution to $R^2$ over all possible orderings in which the three components enter the model. Because averaging over orderings removes the dependence on any single variable ordering, the resulting shares are stable under collinearity and sum exactly to the total model $R^2$. Shares are computed separately for each cohort at weeks 6 and 11 and then averaged. Bivariate Spearman rank correlations from meta-analysis provide a complementary contrast: they show how each component associates with grade in isolation, before removing shared variance.

### 4.3 Heterogeneity across the grade distribution: quantile regression

To test whether engagement matters equally for all students or more strongly for lower-performing students, quantile regression (Koenker and Bassett, 1978) is used. The model takes the form

$$\text{grade}_i = \beta_0(\tau) + \beta_1(\tau)\, z_{\text{IDF},i} + \varepsilon_i(\tau),$$

where $z_{\text{IDF},i}$ is cumulative engagement standardised within cohort and week, $\tau$ ranges over $\{0.1, 0.25, 0.5, 0.75, 0.9\}$, and $\beta_1(\tau)$ gives the grade-point change per one standard deviation of engagement at that quantile of the grade distribution. If engagement matters equally for all students, $\beta_1(\tau)$ should be roughly constant across $\tau$; if it matters more for lower-performing students, the slope at $\tau = 0.1$ should exceed that at $\tau = 0.5$ (the median). The descriptive contrast $\beta_1(0.1) - \beta_1(0.5)$ is supplemented by bootstrap confidence intervals based on 500 student-level resamples per cohort. Indicator-specific quantile models (one of Frequency, Immediacy or Diversity at a time) identify which component drives lower-tail associations. Analyses are conducted at weeks 6 and 11.

### 4.4 Timing and early-course usefulness

Weekly Spearman correlations between cumulative engagement and final grade were tracked across the term to assess how early the association emerges. For each cohort, a bootstrap procedure identified the earliest week at which the correlation was statistically indistinguishable from its end-of-term value, providing a practical sense of when the signal stabilises.

### 4.5 Early-warning performance

Students were grouped into engagement quintiles at selected weeks, and the distribution of final grades within each quintile was examined. The bottom engagement quintile was treated as a risk flag and evaluated using the area under the ROC curve (AUC), recall and precision for identifying students scoring below 50% at the end of term. These complementary measures are reported because AUC alone can look favourable in imbalanced samples while obscuring the proportion of flagged students who ultimately require support (Conijn et al., 2017; Johnston et al., 2025).

### 4.6 Assessment-component validity

Where separate assessment marks were available, engagement was correlated with each component and Steiger's test for dependent correlations was used to compare whether engagement aligns more strongly with individually assessed work (exams, quizzes) than with group- or peer-marked components (Steiger, 1980; Almond, 2009).

### 4.7 Temporal regularity and trajectory analysis

Timing and regularity features were added to models already containing F, I and D to test whether *how* students engage over time adds information beyond *how much* and *how broadly* they engage. Separately, latent-class mixed models were applied to weekly cumulative IDF profiles to identify groups of students following distinct engagement trajectories and to compare their final grades. To assess operational usefulness, end-of-term class labels were compared with early assignments made by matching each student's partial weekly profile (weeks 1–$w$) to the corresponding segment of the end-of-term class centroids. Latent-class mixed models were estimated using the `lcmm` framework (Proust-Lima et al., 2017).

### 4.8 Robustness checks

The main conclusions were examined under degree-programme adjustment, false-discovery-rate correction using the Benjamini–Hochberg procedure across weekly association tests, alternative bounded outcome specifications, and variation in the at-risk engagement threshold. These checks were kept brief and interpreted in the context of the primary findings rather than treated as a separate technical audit.

## 5 Results

Results are reported first for the overall association between cumulative engagement and final grade, then for the component and grade-distribution analyses, before turning to timing, trajectories, assessment alignment, early-warning performance and robustness.

### 5.1 Overall engagement–grade association

Across all five cohorts, cumulative engagement is positively and moderately associated with final grade. The random-effects pooled rank correlation for cumulative IDF at week 11 is $r = 0.32$ (95% CI $[0.25, 0.38]$), with low-to-moderate heterogeneity across cohorts ($I^2 = 39\%$). Frequency, Immediacy and Diversity show similar pooled associations when analysed separately (Table 2). On the grade scale, the mixed-effects estimate corresponds to approximately 3.8 grade points per one standard deviation increase in engagement. Figure 1 shows that every cohort lies above zero, but the association is far from perfect prediction: engagement is a useful but imperfect proxy for academic performance.

**Table 2:** Pooled rank correlations and mixed-effects slopes at week 11.

| Indicator | Pooled $r$ [95% CI] | $I^2$ | Slope (SE) | Moderator $p$ |
|---|---|---|---|---|
| Frequency | 0.28 [0.22, 0.34] | 27% | 3.36 (0.53) | 0.92 |
| Immediacy | 0.30 [0.24, 0.37] | 37% | 3.63 (0.58) | 0.88 |
| Diversity | 0.31 [0.24, 0.37] | 33% | 3.62 (0.53) | 0.87 |
| **IDF** | **0.32 [0.25, 0.38]** | **39%** | **3.79 (0.58)** | 0.89 |

*Source: author's analysis of UCL Moodle log data.*

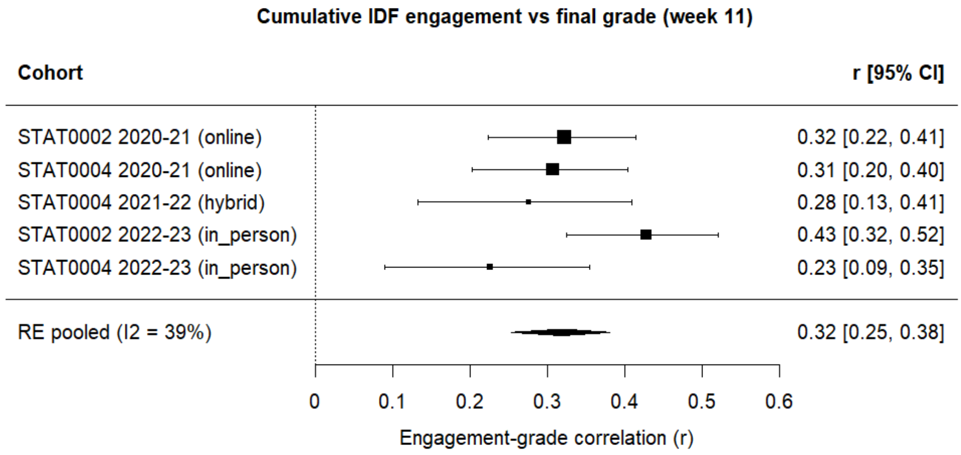



**Fig. 1:** Cohort-specific and pooled week-11 correlations between cumulative IDF engagement and final grade. All cohorts show positive associations; the random-effects estimate summarises the overall moderate engagement–grade relationship.

STAT0002 deliveries show stronger cohort-level associations than STAT0004 ($r$ = 0.28–0.41 versus $r$ = 0.19–0.30), consistent with the module-level patterns noted in Section 3. The pooled estimate nonetheless remains positive and moderate because every cohort contributes in the same direction.

When students are ranked by cumulative engagement and grouped into quintiles, final grades tend to rise across successive quintiles, particularly in STAT0002, where quintile separation is clearest (Figure 2). At weeks 3 and 6, the lowest-engagement quintile is the group whose grade distribution most often approaches or crosses the 50% low-performance threshold, while the highest-engagement quintile consistently shows the strongest outcomes. Separation between adjacent middle quintiles is weaker, which is consistent with a monotone but noisy relationship rather than sharp boundaries between moderate engagement levels. Together, the pooled correlation, positive cohort-level associations and quintile patterns support the conclusion that the metric *travels* to these new module-year contexts with a moderate, robust association.

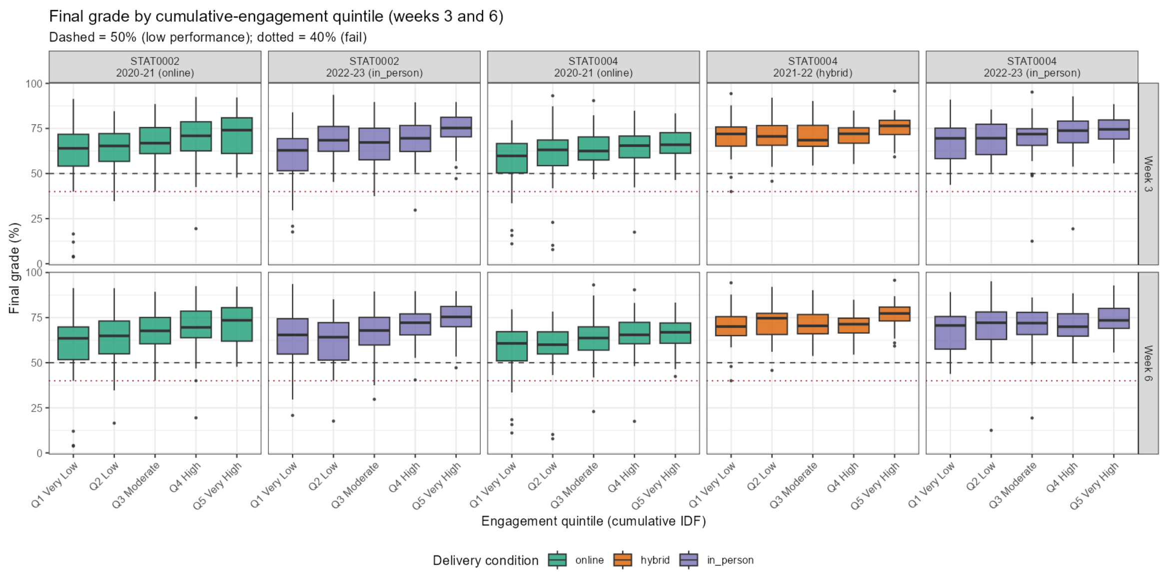



**Fig. 2:** Final-grade distributions by engagement quintile at weeks 3 and 6.

## 5.2 Component importance

Treating engagement as a single composite score confirms that it relates to grades, but obscures *which* learning behaviours carry unique predictive information. Frequency (how often students interact), Diversity (the variety of resource types accessed) and Immediacy (how promptly students engage with newly released content) are strongly correlated within cohorts: mean pairwise $|r| \approx 0.75$–$0.83$, with variance inflation factors between 2.6 and 4.4 at weeks 6 and 11. Under such collinearity, ordinary multiple-regression coefficients are unstable and can even change sign across cohorts; a method that partitions explained variance fairly is required.

**Bivariate contrast.** Bivariate pooled rank correlations at week 11 show similar associations for all three indicators when analysed separately: $r = 0.28$ (Frequency), $r = 0.30$ (Immediacy) and $r = 0.31$ (Diversity; Table 2). Taken alone, this would suggest that all components matter equally. The LMG results show that this is misleading once shared variance is removed.

**End-of-term results.** At end-of-term (week 11), Diversity accounts for the largest mean share of explained variance (37%), Immediacy is intermediate (36%), and Frequency the smallest (27%). Frequency has the smallest LMG share in every cohort (Table 3, Figure 3). Cohort-level patterns vary: Immediacy dominates in STAT0002 online (45%), whereas Diversity dominates in STAT0004 online and in-person (45% and 42% respectively).

**Table 3:** LMG relative-importance shares (%) at week 11.

| Cohort | Frequency | Immediacy | Diversity |
|---|---|---|---|
| STAT0002 online | 28 | 45 | 27 |
| STAT0002 in-person | 28 | 36 | 36 |
| STAT0004 online | 23 | 32 | 45 |
| STAT0004 hybrid | 31 | 32 | 38 |
| STAT0004 in-person | 25 | 33 | 42 |
| **Mean** | **27** | **36** | **37** |

*Source: author's analysis of UCL Moodle log data.*

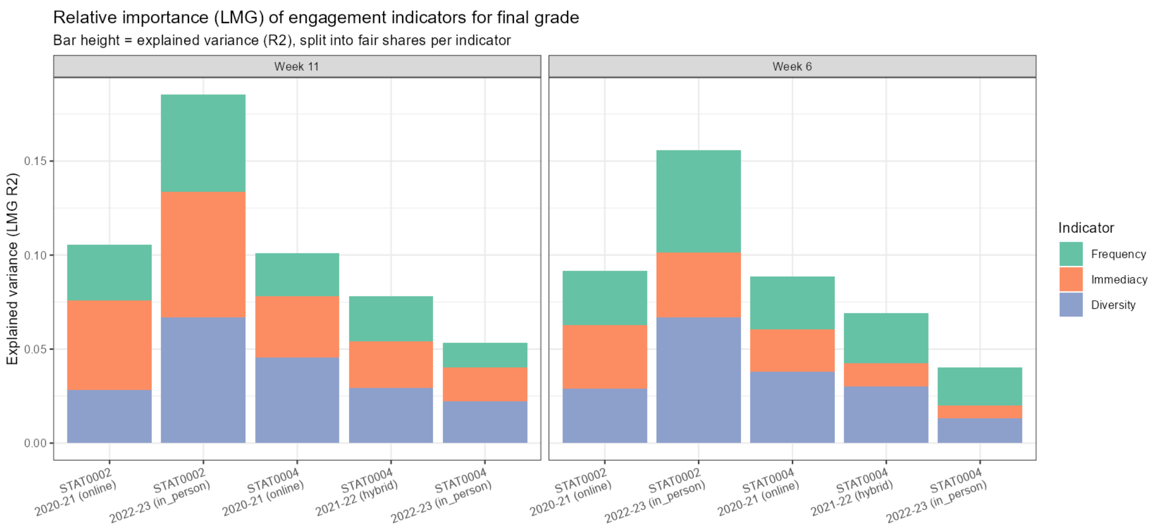



**Fig. 3:** LMG relative-importance shares for Frequency, Immediacy and Diversity by cohort at weeks 6 (left) and 11 (right). Diversity and Immediacy typically account for more unique explained variance than Frequency.

**Mid-term results.** The same decomposition at week 6, the course midpoint and a common decision point for early support, shows a broadly similar ranking: Diversity remains the top contributor on average (39%), with Immediacy lowest (24%) and Frequency intermediate (37%). One cohort (STAT0004 in-person) assigns a larger week 6 share to Frequency (50%), but Frequency never ranks first at week 11 in any cohort. The pooled pattern supports interpreting breadth of resource use as the most consistent unique signal, with promptness also carrying substantial information, especially in first-year STAT0002.

**Table 4:** LMG relative-importance shares (%) at week 6.

| Cohort | Frequency | Immediacy | Diversity |
|---|---|---|---|
| STAT0002 online | 31 | 37 | 32 |
| STAT0002 in-person | 35 | 22 | 43 |
| STAT0004 online | 32 | 25 | 43 |
| STAT0004 hybrid | 38 | 18 | 43 |
| STAT0004 in-person | 50 | 17 | 33 |
| **Mean** | **37** | **24** | **39** |

*Source: author's analysis of UCL Moodle log data.*

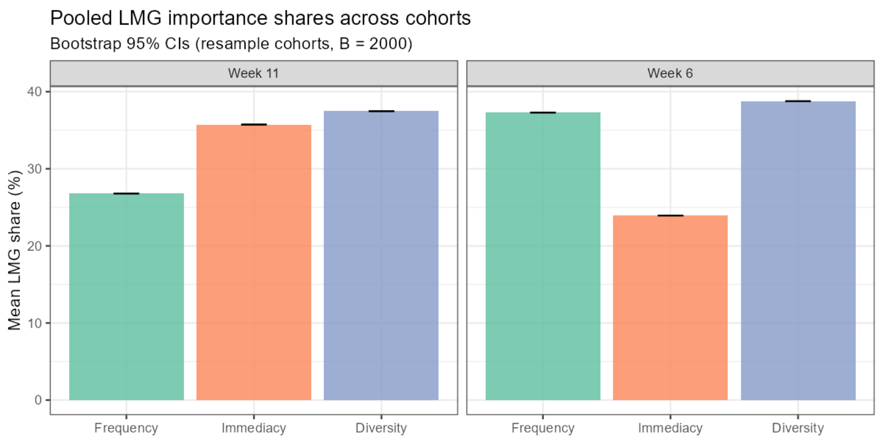



**Fig. 4:** Pooled mean LMG importance shares across cohorts at weeks 6 and 11.

## 5.3 Variation across the grade distribution

A single average association, whether from OLS or Spearman correlation, can hide whether engagement is more consequential for students already at academic risk. Quantile regression addresses this by estimating separate slopes at different points in the grade distribution.

**Composite engagement.** At week 11, the IDF slope is steeper at $\tau = 0.1$ than at $\tau = 0.5$ in **every cohort** (Table 5). On average, the lower-tail slope exceeds the median slope by 2.3 grade points per standard deviation of engagement. Slopes generally decrease from $\tau = 0.1$ through $\tau = 0.5$ to $\tau = 0.9$, indicating that engagement variation is most consequential among lower-performing students. For example, in STAT0002 in-person the slope is 7.9 points at $\tau = 0.1$ versus 5.0 at the median and 3.0 at $\tau = 0.9$. STAT0004 hybrid shows the smallest tail excess (0.6 points), suggesting cohort-specific heterogeneity in how sharply the lower tail diverges.

Student-level bootstrap confidence intervals for $\beta_1(0.1) - \beta_1(0.5)$ confirm the pattern in two of five cohorts at week 11 (STAT0002 and STAT0004 in-person); in the remaining cohorts the point estimates are positive but intervals are wide (Table 6). At week 6, three cohorts show positive descriptive contrasts and one (STAT0004 in-person) has a bootstrap interval excluding zero.

**Table 5:** Quantile-regression slopes for cumulative IDF at week 11 (grade points per +1 SD).

| Cohort | $\tau = 0.1$ | $\tau = 0.5$ | $\tau = 0.9$ | $\tau_{0.1} - \tau_{0.5}$ |
|---|---|---|---|---|
| STAT0002 online | 5.32 | 3.22 | 2.58 | 2.11 |
| STAT0002 in-person | 7.88 | 5.04 | 2.96 | 2.84 |
| STAT0004 online | 6.06 | 3.15 | 2.17 | 2.91 |
| STAT0004 hybrid | 3.26 | 2.66 | 1.34 | 0.59 |
| STAT0004 in-person | 3.97 | 1.10 | 1.28 | 2.88 |

*Source: author's analysis of UCL Moodle log data.*

**Table 6:** Bootstrap 95% CI for $\tau_{0.1} - \tau_{0.5}$ contrast (week 11).

| Cohort | Contrast | 95% CI | CI > 0? |
|---|---|---|---|
| STAT0002 online | 2.11 | [−1.60, 4.79] | No |
| STAT0002 in-person | 2.84 | [0.16, 5.54] | Yes |
| STAT0004 online | 2.91 | [−0.85, 4.77] | No |
| STAT0004 hybrid | 0.59 | [−2.62, 4.49] | No |
| STAT0004 in-person | 2.88 | [0.26, 5.77] | Yes |

*Source: author's analysis of UCL Moodle log data.*

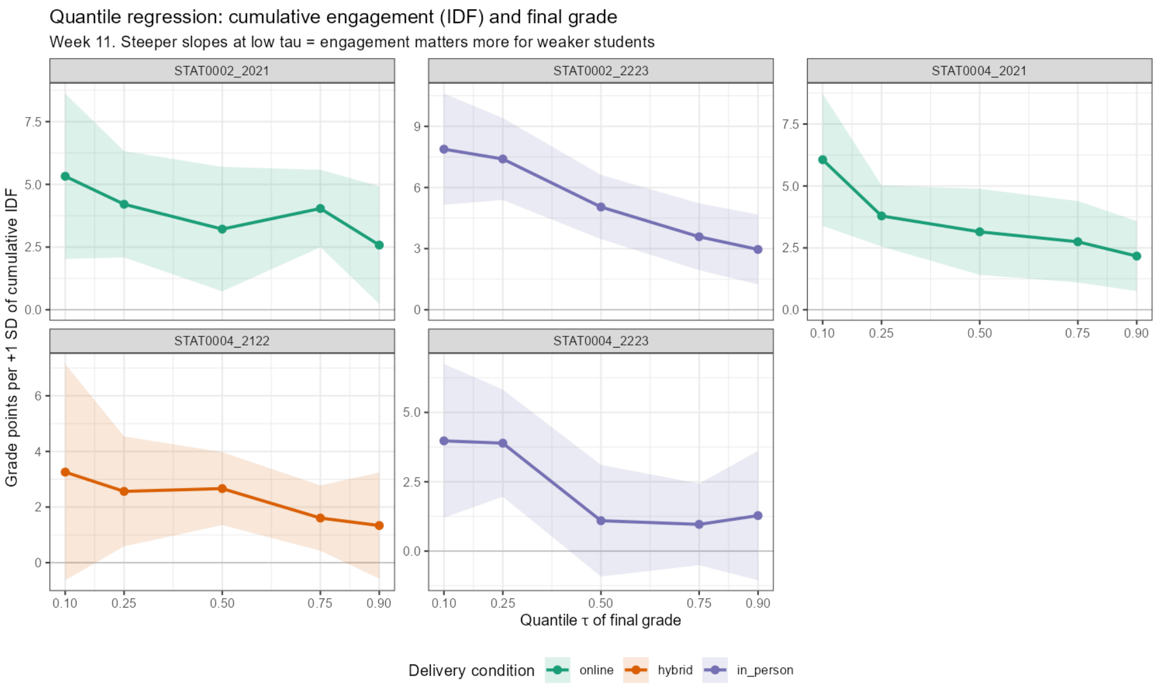



**Fig. 5:** Quantile-regression slopes of final grade on cumulative IDF at week 11, with 95% confidence intervals, by cohort.

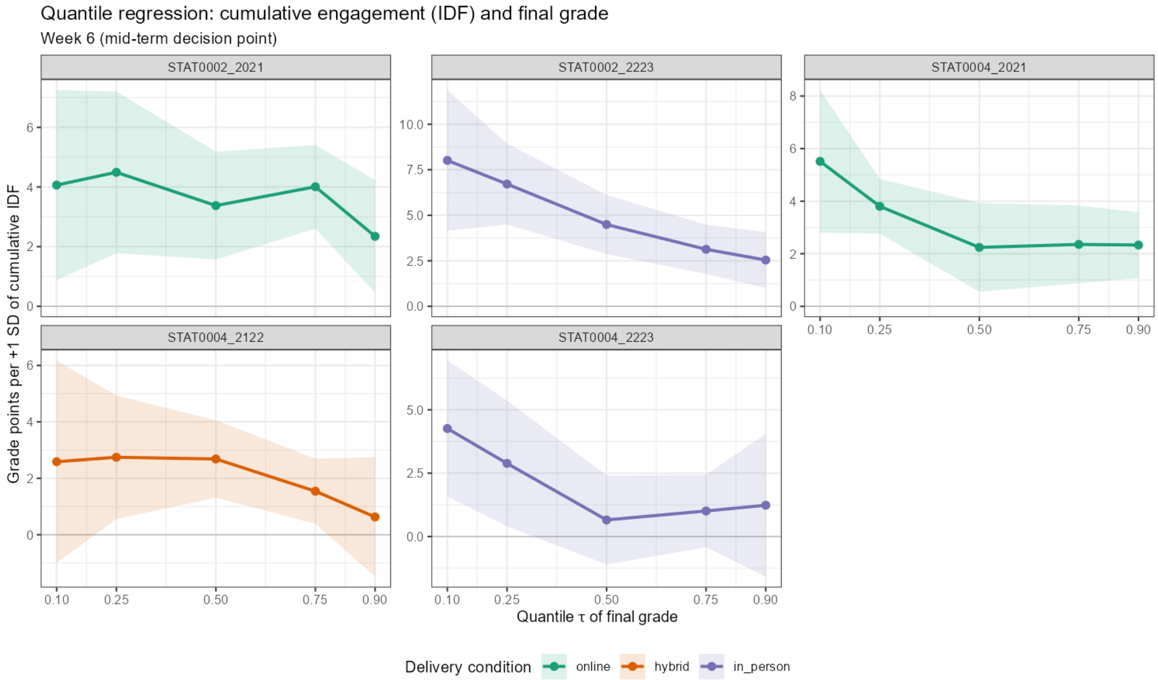



**Fig. 6:** Quantile-regression slopes at week 6 (mid-term). The lower-tail pattern is visible in most cohorts by the course midpoint.

**Indicator-specific associations.** Univariate quantile regression (one indicator at a time) shows that **Immediacy** has the largest mean slope at $\tau = 0.1$ across cohorts at week 11 (4.9 grade points per SD), followed by Diversity and Frequency (Figure 7). This complements the LMG finding that Diversity carries the most unique variance in multivariate models: promptness appears especially salient for distinguishing outcomes among lower-performing students when indicators are considered separately.

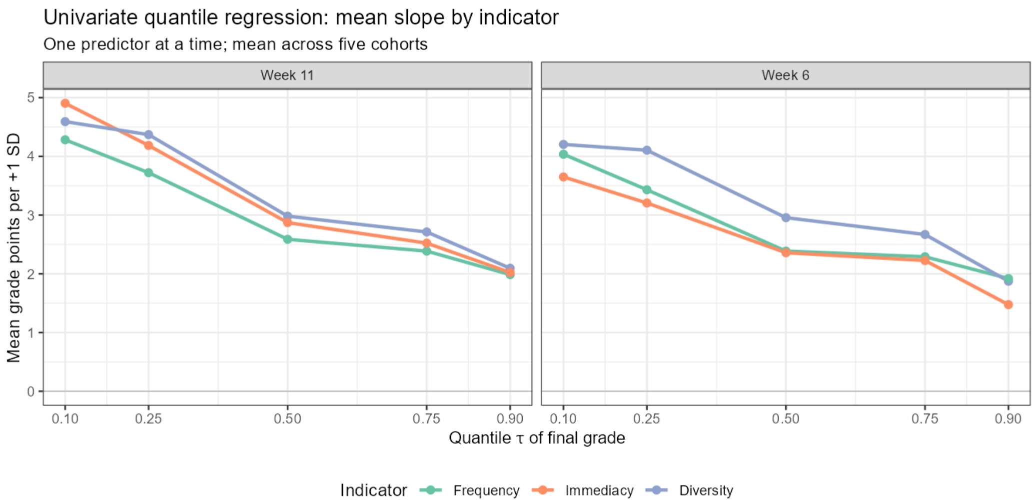



**Fig. 7:** Mean univariate quantile-regression slopes by indicator (weeks 6 and 11; mean across five cohorts). Larger coefficients at low $\tau$ indicate stronger associations among lower-performing students.

## 5.4 Timing and stabilisation

Bootstrap stabilisation analysis asks whether additional weeks materially change the engagement–grade correlation, not whether the correlation is already strong (Table 7, Figure 8). In four cohorts, the weekly correlation is statistically close to its end-of-term value from week 1 or 2, but this sometimes reflects a relatively flat and modest signal rather than impressive early predictive power. For example, STAT0004 in-person stabilises early yet has a weak end-of-term association ($r = 0.19$); STAT0004 hybrid similarly stabilises at week 1 with $r = 0.23$. By contrast, STAT0002 in-person shows a strengthening association that does not stabilise until around week 9, reaching the strongest cohort-level correlation ($r \approx 0.41$), suggesting that later engagement adds meaningful information in that delivery. Stabilisation timing and association strength should therefore be read separately: early stabilisation does not guarantee a strong early-warning signal.

**Table 7:** Stabilisation week and end-of-term IDF–grade correlation.

| Cohort | Stabilisation week | End-of-term $r$ |
|---|---|---|
| STAT0002 online | 2 | 0.28 |
| STAT0002 in-person | 9 | 0.41 |
| STAT0004 online | 1 | 0.30 |
| STAT0004 hybrid | 1 | 0.23 |
| STAT0004 in-person | 1 | 0.19 |

*Source: author's analysis of UCL Moodle log data.*

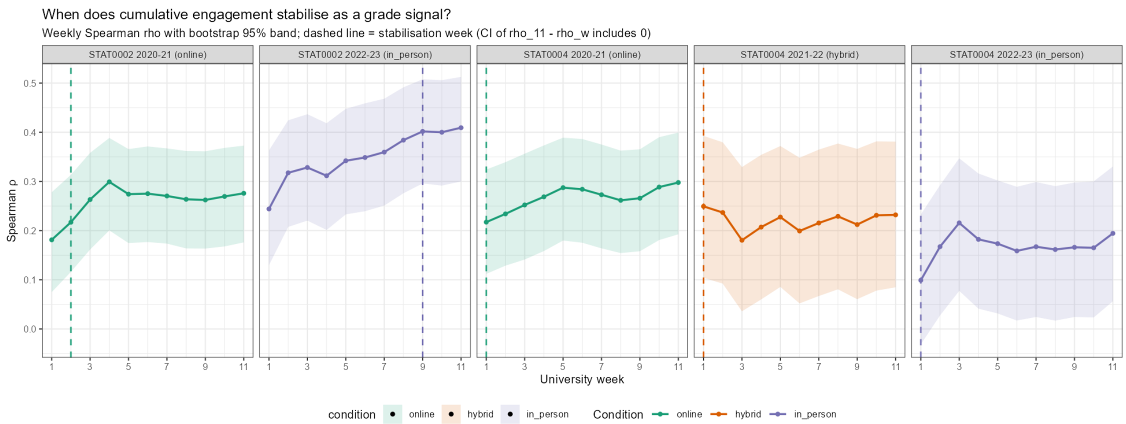



**Fig. 8:** Weekly Spearman correlation between cumulative engagement and final grade by cohort, with bootstrap stabilisation week marked. Early stabilisation can reflect a flat, modest signal as well as a strong one; compare stabilisation week with end-of-term $r$ in Table 7.

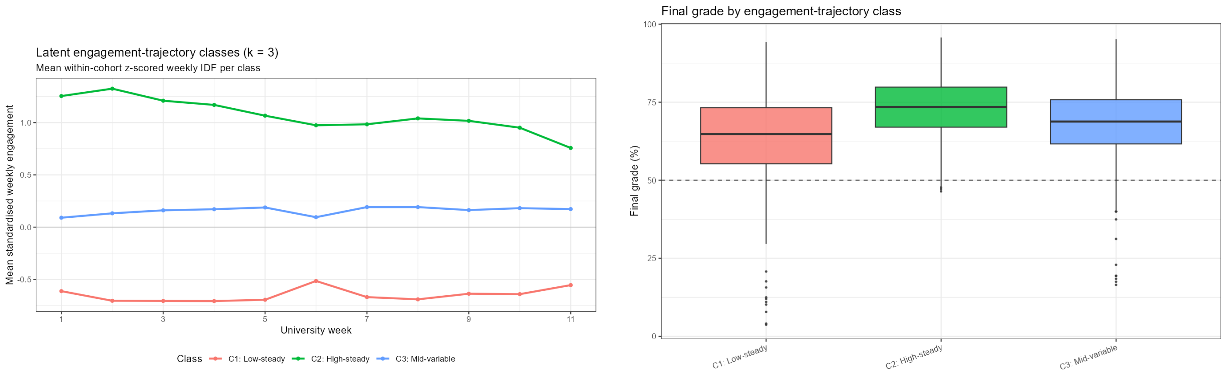
## 5.5 Engagement trajectories

Latent-class analysis identifies three weekly cumulative IDF trajectory types: low-steady, high-steady and mid-variable (Figure 9). Final grades differ markedly across classes (Kruskal–Wallis $p = 2.6 \times 10^{-17}$). The high-steady class averages 72.9, compared with 63.2 for the low-steady class: a gap of about 9.7 grade points. The low-steady class also contains the highest share of low performers (15.7%). Sustained engagement *patterns* over the term therefore separate outcomes more sharply than a single cumulative score at one time point, and identify a group whose profile is persistently low rather than temporarily quiet.

**Table 8:** Trajectory classes and grade outcomes ($N = 1{,}290$).

| Class | $N$ | Mean grade | % < 50 |
|---|---|---|---|
| Low-steady | 439 | 63.2 | 15.7 |
| High-steady | 165 | 72.9 | 2.4 |
| Mid-variable | 686 | 68.3 | 5.5 |

*Source: author's analysis of UCL Moodle log data.*




**(a)** Trajectory shapes

**(b)** Grades by class

**Fig. 9:** Latent engagement trajectories (left) and final-grade distributions by trajectory class (right). Three trajectory types separate outcomes by roughly 9.7 grade points between the highest- and lowest-engagement profiles.

A natural operational question is how early a student can be assigned to a trajectory class with reasonable confidence. End-of-term class labels were therefore compared with early assignments made by matching each student's partial weekly profile (weeks 1–$w$) to the corresponding segment of the end-of-term class centroids (Table 9, Figure 10). Overall agreement rises from 71% at week 3 to 90% by week 9. For the low-steady class specifically, recall reaches 81% by week 4 and 88% by week 6; precision that an early low-steady flag corresponds to the true end-of-term low-steady class is 77% at week 4 and 81% at week 6. These levels are materially higher than the bottom-quintile flag's precision for identifying students who ultimately score below 50% (about 15–26%). However, among students flagged as low-steady early, only about 15% still score below 50%, similar to the quintile rule. Trajectory class is therefore a more reliable early label for *sustained low engagement* than the quintile is for *eventual failure*: operationally useful for targeted, pattern-based outreach from mid-term onwards, but not a sharp failure predictor.

**Table 9:** Early recovery of end-of-term trajectory class from partial weekly profiles.

| Week | Agreement (%) | Low-steady recall (%) | Low-steady precision (%) | % flagged < 50 |
|---|---|---|---|---|
| 3 | 70.9 | 78.6 | 69.0 | 14.8 |
| 4 | 76.9 | 81.3 | 76.8 | 14.6 |
| 6 | 81.6 | 88.2 | 80.8 | 14.8 |

*Source: author's analysis of UCL Moodle log data.*

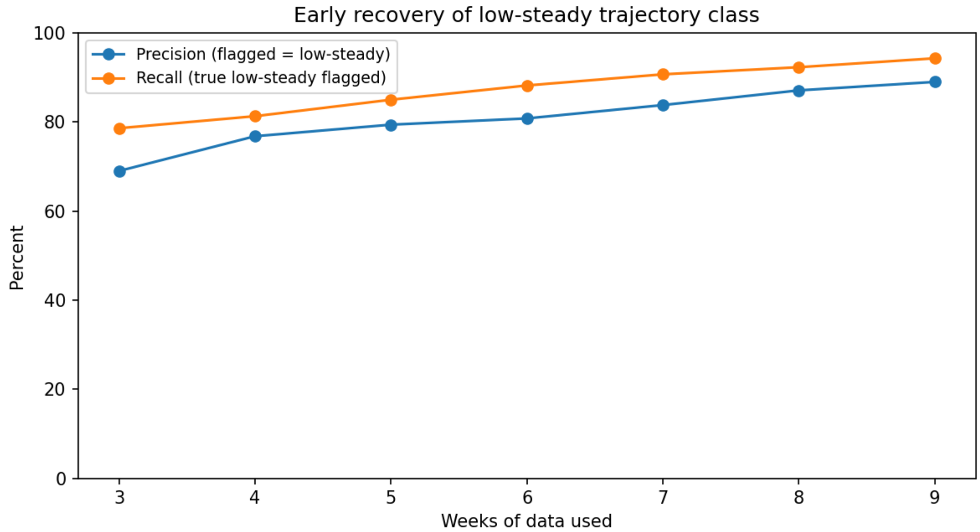



**Fig. 10:** Recall and precision for early assignment to the low-steady trajectory class as more weekly data accumulate. Agreement improves steadily from week 3 onwards.

### 5.6 Assessment alignment and temporal regularity

Engagement correlates more strongly with individually assessed components than with group- or peer-marked work where both are available (Figure 11). In STAT0002 in-person, the correlation with the exam grade ($r = 0.40$) is significantly larger than with peer-marking ($r = 0.20$; Steiger $z = 2.98$, $p = 0.003$). Other component contrasts point in the same direction but are not statistically significant. This supports the interpretation that Moodle engagement reflects individual study behaviour more faithfully than outcomes shaped by group dynamics or peer assessment.

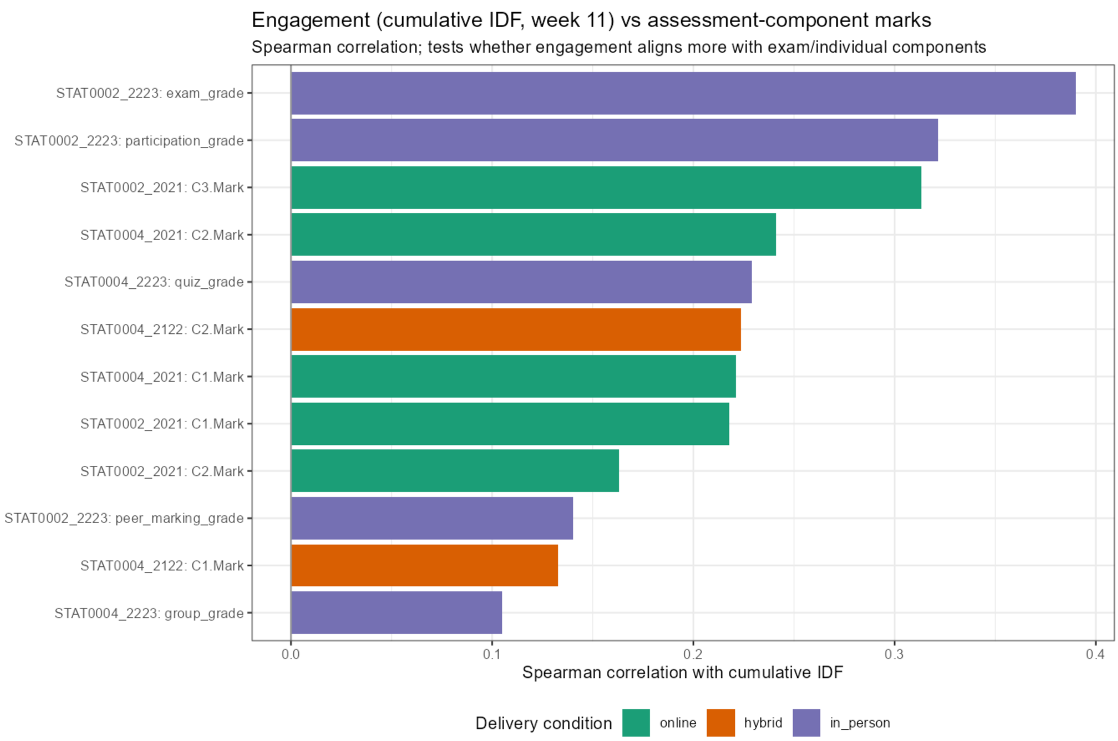



**Fig. 11:** Engagement–assessment component correlations by cohort.

Timing and regularity features add incremental explanatory power beyond F, I and D in the STAT0002 cohorts, where engagement showed stronger alignment with individual assessment components (Table 10, Figure 12). At week 6, the gain in $R^2$ is about 0.05 in both STAT0002 deliveries ($p = 0.003$ and $p = 0.006$), but not statistically significant in the STAT0004 deliveries. How regularly students work carries additional diagnostic information in STAT0002, beyond what the core composite already captures.

**Table 10:** Incremental $R^2$ from temporal features at week 6.

| Cohort | $R^2$ (F/I/D) | $R^2$ (full) | $\Delta R^2$ | $p$ |
|---|---|---|---|---|
| STAT0002 online | 0.091 | 0.143 | 0.052 | 0.003 |
| STAT0002 in-person | 0.156 | 0.213 | 0.057 | 0.006 |
| STAT0004 online | 0.089 | 0.094 | 0.005 | 0.939 |
| STAT0004 hybrid | 0.069 | 0.102 | 0.033 | 0.424 |
| STAT0004 in-person | 0.040 | 0.094 | 0.053 | 0.094 |

*Source: author's analysis of UCL Moodle log data.*

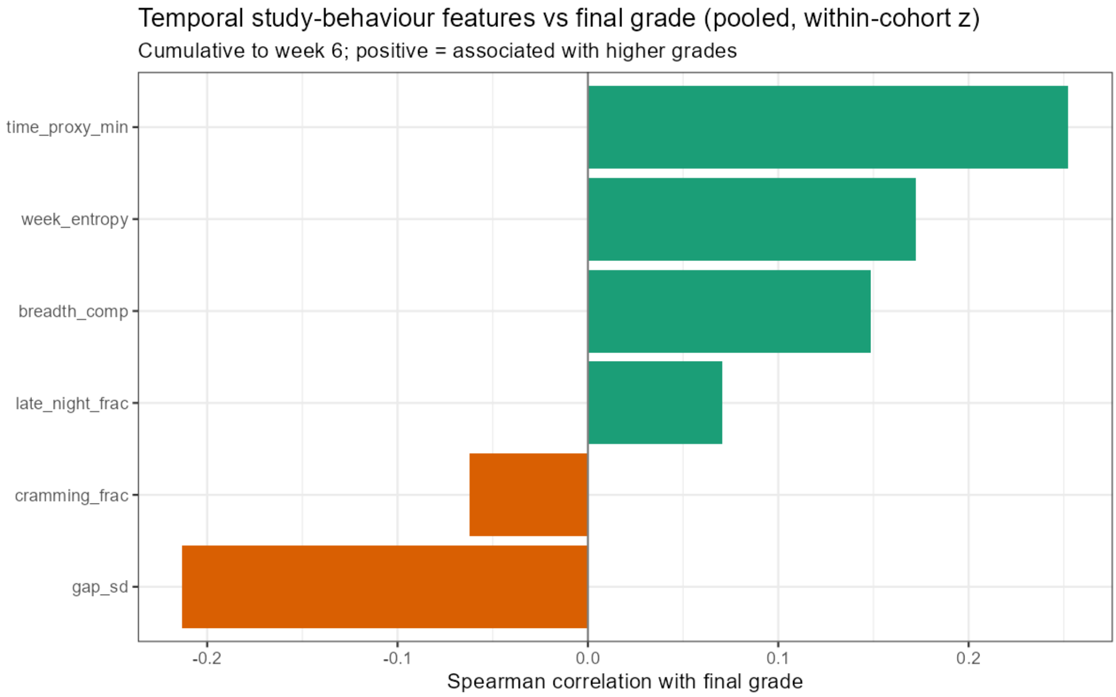



**Fig. 12:** Temporal and regularity features correlated with final grade.

The diagnostic extensions are not uniform across modules. Temporal regularity adds significant incremental $R^2$ only in STAT0002, where engagement also aligns more strongly with individually assessed components. In STAT0004, where associations are weaker and group or coursework components carry more weight, the composite and its extensions explain less outcome variance, consistent with greater off-platform study, second-year autonomy, or looser chapter–assessment alignment weakening the Moodle signal rather than invalidating the metric outright.

## 5.7 Early-warning performance

Although the association is stronger in the lower grade tail, operational flagging still faces modest precision. At the course midpoint, the bottom engagement quintile in STAT0002 identifies roughly 38–47% of students who ultimately score below 50% (AUC around 0.71–0.76), but precision is only about 15–26%: roughly three-quarters to four-fifths of flagged students may still pass. Figure 13 illustrates this recall–precision trade-off. The metric is therefore useful for identifying students with low *observed* Moodle engagement, not for identifying students who will fail. Appropriate responses are broad, low-cost and supportive rather than intensive or punitive.

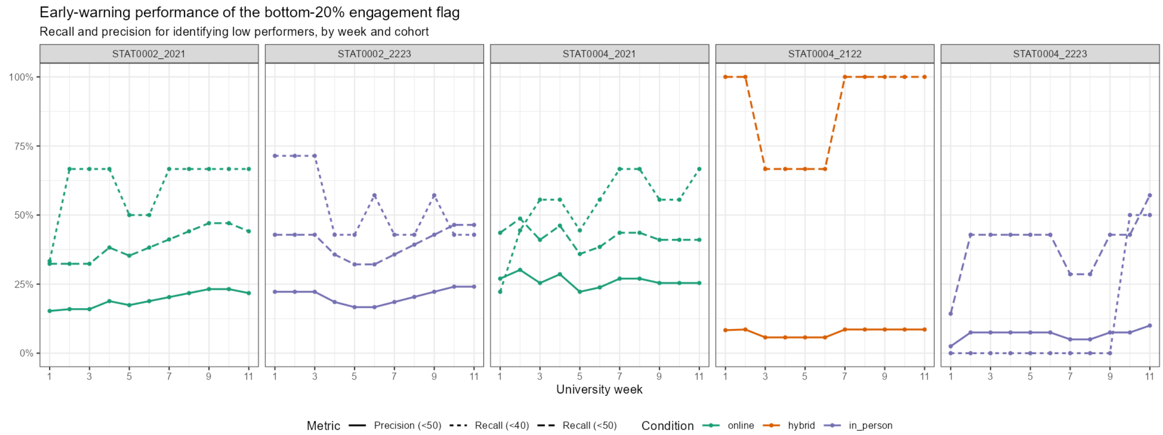



**Fig. 13:** Recall and precision for the bottom engagement quintile across weeks. Modest precision implies that most flagged students may still achieve passing grades.

## 5.8 Boundary condition: delivery period and cohort confounding

Because delivery condition is inseparable from cohort and year in this dataset, comparisons across online, hybrid and in-person periods are best treated as a descriptive boundary check rather than a test of delivery effects. Meta-analytic moderator tests do not suggest that delivery label explains between-cohort variation (all $p > 0.87$; Table 2), and weekly correlations appear broadly overlapping across periods (Figure 14). This should not be interpreted as evidence that delivery mode is irrelevant. Module design, assessment structure, cohort composition and Moodle resource organisation may be at least as important as the delivery label itself.

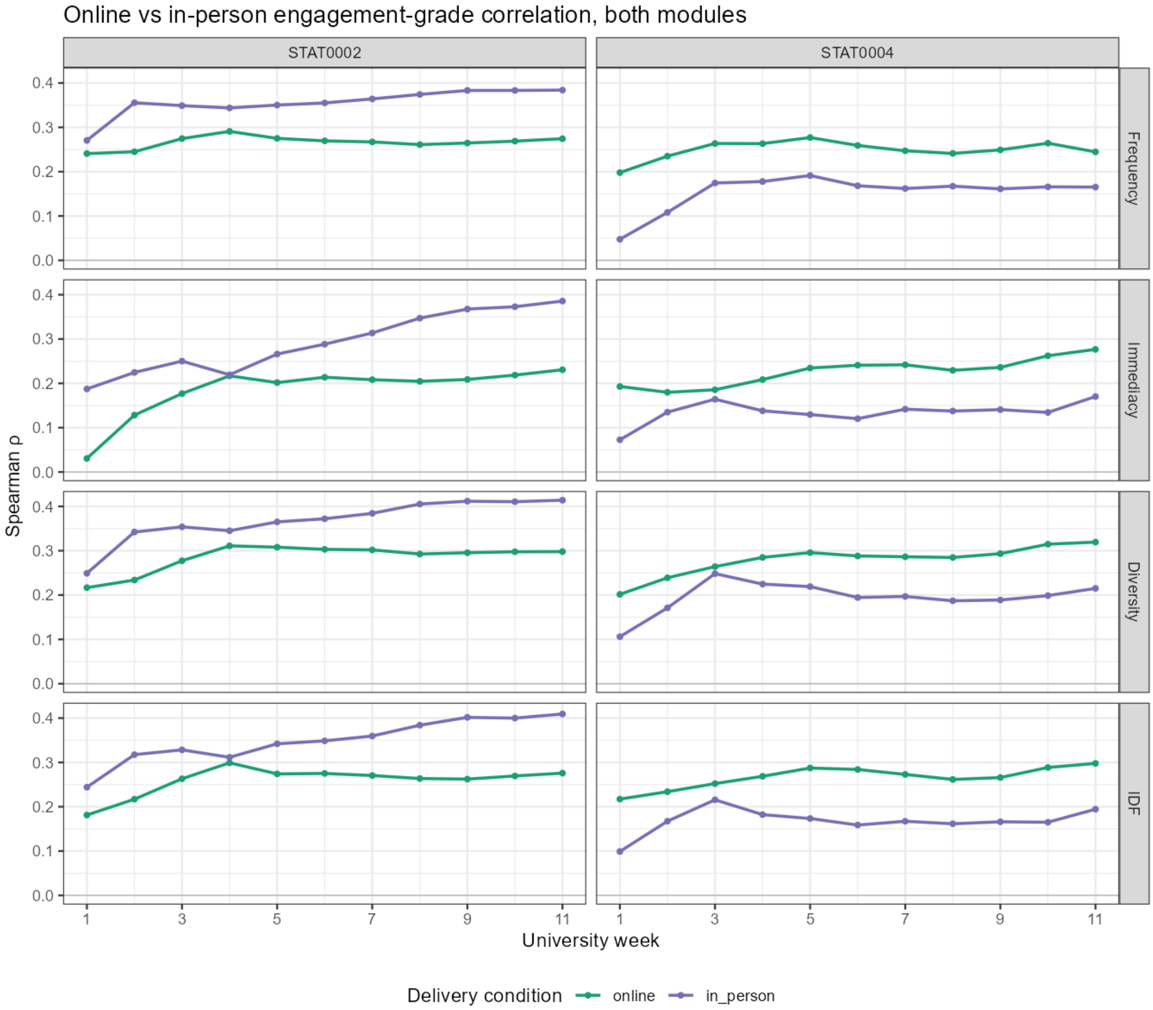



**Fig. 14:** Weekly engagement–grade correlation by module and delivery period. Comparisons are descriptive; condition is confounded with academic year.

### 5.9 Robustness checks

Each check addressed a specific threat to the headline conclusions. Programme adjustment tests whether the pooled engagement slope reflects degree-programme composition rather than study behaviour itself; the slope changes by at most 14%, so programme mix is not driving the result. False-discovery-rate correction using the Benjamini–Hochberg procedure guards against spurious weekly associations from repeated testing across weeks and indicators; 205 of 220 tests remain significant. Alternative bounded-outcome specifications address skew and ceiling effects in grade distributions and agree in sign with the main models in all five cohorts. Finally, varying the at-risk engagement threshold examines whether early-warning conclusions depend on an arbitrary quintile cut-off; recall and precision shift in the expected direction without changing the overall interpretation.

## 6 Discussion

### 6.1 Interpretation of the component and distributional findings

The central contribution of this report is not that Moodle activity predicts grades perfectly: it does not. Rather, it shows that a transparent, chapter-aligned summary of observed engagement retains a moderate relationship with final performance in cohorts independent of the original validation. The pooled association ($r = 0.32$) and positive direction in every cohort extend the evidence of Johnston et al. (2025), while also illustrating a central caution in learning analytics: trace data are indicators of observable behaviour, not direct measures of learning, motivation or effort (Winne, 2020; Motz et al., 2019).

Once the overlap among Frequency, Immediacy and Diversity is taken into account, Diversity and Immediacy carry more unique information about final grades than Frequency. This does not imply that session frequency is unimportant. Instead, it suggests that repeated access alone is an incomplete account of productive online study. Breadth of resource use is consistent with curriculum coverage, while prompt first access is consistent with keeping pace with the intended teaching sequence. This interpretation accords with work showing that frequency, immediacy and regularity capture complementary aspects of online engagement (Hoffman et al., 2023; You, 2016). The contrast between similar bivariate correlations and unequal relative-importance shares also demonstrates why component interpretation should not rely only on individual correlations.

The quantile results add a distributional qualification. Engagement is associated with grade throughout the cohort, but the association is larger among students in the lower tail of the grade distribution. This is an empirical finding of the present analysis rather than evidence that increasing Moodle engagement would cause grades to rise by the estimated amount. It nevertheless supports treating low observed engagement as one input to supportive outreach for students who may be academically vulnerable, rather than as a tool for ranking all students with the same confidence.

## 6.2 Portability, trajectories and diagnostic value

The metric performs differently across the two modules. The stronger engagement–grade associations in STAT0002 than in STAT0004 are plausible given differences in assessment structure, student autonomy and the extent to which study activity occurs outside Moodle. Such contextual variation is expected: the validity and utility of activity logs depend on instructional conditions and course design (Gašević et al., 2016; Motz et al., 2019). It reinforces the need to validate the metric locally rather than assume that a threshold or effect size transfers unchanged to every module.

The extension analyses clarify what is gained by retaining the temporal structure of engagement. Low-steady, mid-variable and high-steady trajectories separate final grades more sharply than a single score at one time point, in line with evidence that longitudinal engagement profiles can reveal meaningful differences in learning outcomes (Saqr and López-Pernas, 2021; Cerezo et al., 2016). Regularity features add information in STAT0002 but not in STAT0004, which is informative rather than a failure: it indicates that temporal patterns should be treated as context-specific diagnostic features. Similarly, stronger correlations with individually assessed components than with peer- or group-marked outcomes are consistent with the fact that group marks do not map cleanly onto individual study behaviour (Almond, 2009).

## 6.3 Implications for early support

The bottom-engagement flag identifies students with low *observed* Moodle engagement, not students who will fail. With precision of about 15–26% at the course midpoint in STAT0002, most flagged students may still pass. This trade-off is common in early-warning systems and should determine the intensity of any response (Conijn et al., 2017; Akçapınar et al., 2019). Appropriate uses are broad, low-cost actions such as study reminders, non-accusatory check-ins, invitations to office hours, human review, or dashboard-level monitoring of chapters and resources. Engagement should be combined with coursework information, attendance where available, tutor knowledge and student self-report before substantive action.

The component and trajectory results make this approach more informative than a single risk label. A student may be behind the release schedule (low Immediacy), using only a narrow range of materials (low Diversity), engaging irregularly, or following a persistently low trajectory. From week 4 onward, the trajectory analysis can identify a sustained low-engagement pattern with substantially higher precision than the bottom-quintile rule identifies eventual low performance. These distinctions can guide supportive, proportionate outreach. They do not justify punitive messaging, high-stakes ranking, or an assumption that low Moodle activity means low effort.

### 6.4 Limitations

The findings are associational and their scope is limited. Delivery condition is perfectly confounded with academic year and cohort, so the data cannot identify causal effects of online, hybrid or in-person delivery. The analysis covers two statistics modules and five deliveries; external validity therefore remains to be tested across disciplines, assessment designs and institutional contexts. Moodle logs capture online behavioural engagement only, leaving offline study, motivation, cognitive engagement and accessibility constraints unobserved (Martin and Borup, 2022; Nkomo et al., 2021). Final grades are imperfect outcome measures, especially when group, peer or non-invigilated assessment contributes. Small numbers of low-performing students in some cohorts also make early-warning estimates unstable. Finally, the metric depends on reliable chapter labelling and resource organisation. Any operational deployment should therefore include local validation of data quality, association strength and intervention thresholds.

## 7 Conclusion

This report validates an existing real-time Moodle engagement metric on new STAT0002 and STAT0004 cohorts spanning online, hybrid and in-person delivery periods. Cumulative engagement is moderately associated with final grades ($r = 0.32$ pooled). Two findings receive particular emphasis. **First**, decomposing engagement into Frequency, Immediacy and Diversity shows that breadth and promptness of resource use outweigh raw session frequency once indicator overlap is removed, a distinction that bivariate correlations alone would not reveal. **Second**, quantile regression shows that engagement matters more for lower-performing students than for the average student, supporting targeted support for at-risk learners rather than uniform prediction across the grade distribution. Trajectory classes, temporal regularity and assessment-component alignment provide additional diagnostic context. Delivery comparisons are descriptive only. The metric should be used for low-cost supportive monitoring with local validation, not high-stakes prediction.

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